SYLLABUS

UNIT-I: Data Management: Design Data Architecture and manage the data for analysis, understand various sources of Data like Sensors/Signals/GPS etc. Data Management, Data Quality (noise, outliers, missing values, duplicate data) and Data Processing & Processing.

UNIT-II: Data Analytics: Introduction to Analytics, Introduction to Tools and Environment, Application of Modeling in Business, Databases & Types of Data and variables, Data Modeling Techniques, Missing Imputations etc. Need for Business Modeling.

UNIT-III: Regression – Concepts, Blue property assumptions, Least Square Estimation, Variable Rationalization, and Model Building etc. Logistic Regression: Model Theory, Model fit Statistics, Model Construction, Analytics applications to various Business Domains etc.

UNIT-IV: Object Segmentation: Regression Vs Segmentation – Supervised and Unsupervised Learning, Tree Building – Regression, Classification, Overfitting, Pruning and Complexity, Multiple Decision Trees etc. Time Series Methods: Arima, Measures of Forecast Accuracy, STL approach, Extract features from generated model as Height, Average Energy etc and Analyse for prediction.

UNIT-V: *Data Visualization: Pixel-Oriented Visualization Techniques, Geometric Projection Visualization Techniques, Icon-Based Visualization Techniques, Hierarchical Visualization Techniques, Visualizing Complex Data and Relations.*

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Data Visualization

Data visualization: Data visualization is the graphical representation of information and data. By using visual elements like charts, graphs, and maps, data visualization tools provide an accessible way to see and understand trends, outliers, and patterns in data.

Uses of data visualization

- Powerful way to explore data with presentable results.
- Primary use is the preprocessing portion of the data mining process.
- Supports in data cleaning process by finding incorrect and missing values.

General Data visualization Techniques:

- Box plots
- Histograms
- Heat maps
- Charts
- Tree maps

Data visualization is one of the steps of the data science process, which states that after data has been collected, processed and modeled, it must be visualized for conclusions to be made.

Data visualization Techniques:

Data visualization aims to communicate data clearly and effectively through graphical representation. Data visualization has been used extensively in many applications for example, at work for reporting, managing business operations, and tracking progress of tasks.

More popularly, we can take advantage of visualization techniques to discover data relationships that are otherwise not easily observable by looking at the raw data.

Data visualization Techniques:

1. Pixel Oriented Visualization technique
2. Geometric Projection Visualization technique-
   a. Scatter plot matrices
   b. Hyper slice
   c. Parallel coordinates
3. Icon based visualization techniques
4. Hierarchical Visualization techniques.

Pixel-Oriented Visualization Techniques:

- A simple way to visualize the **value of a dimension** is to represent each value using a **pixel**, where the **color of the pixel corresponds to the value**.
- For a dataset containing **m dimensions**, pixel-oriented techniques create **m separate windows** on the screen, with one window representing each dimension.
- The **m-dimensional values of each data record** are mapped to **m pixels** at the corresponding positions in these windows. The **color of each pixel represents the value** of that dimension.
- Within each window, the data values are arranged according to a **common global order** that is shared by all the windows.
- This **global order** can be obtained by sorting all data records according to a criterion that is **meaningful for the particular analysis or task**.

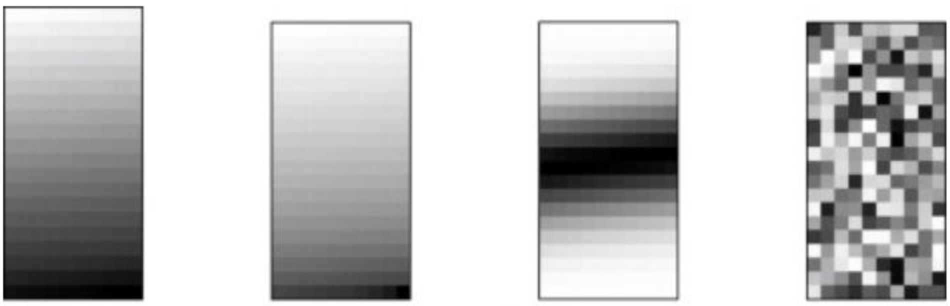
Pixel-based visualizations use that approach and are capable of displaying large amounts of data on a single screen.

Case Study:

All Electronics maintains a customer information table with 4 dimensions:

- Income
- Transaction Volume
- Credit Limit
- Age

The objective is to analyze the correlation between Income and the other attributes using pixel-oriented visualization.



(a) *income* (b) *credit_limit* (c) *transaction_volume* (d) *age*

Geometric Projection Visualization Techniques

A major drawback of **pixel-oriented visualization techniques** is that they are not very effective for understanding the **distribution of data in a multidimensional space**.

Geometric projection techniques overcome this limitation by projecting multidimensional data into a lower-dimensional space, such as **2D or 3D**, making the data easier to analyze and interpret.

These techniques are useful for identifying **outliers, clusters, patterns, and correlations between attributes** in multivariate datasets. They achieve this by applying various **transformations and projections** to the data.

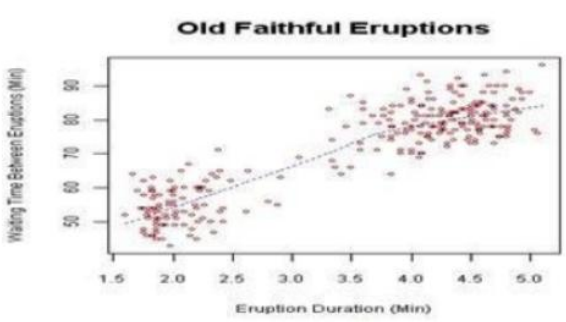
When dealing with **large datasets**, a clustering algorithm is often applied before visualization. This helps reduce **overlapping, clutter, and visual complexity**, resulting in a clearer representation of the data.

Some commonly used geometric projection techniques include:

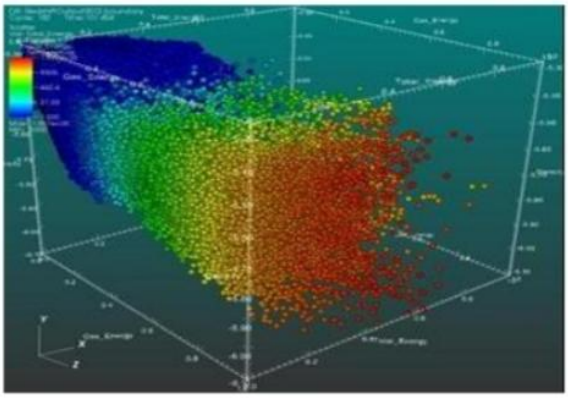
Scatter Plots

A **scatter plot** is one of the most widely used geometric projection visualization techniques. It can be represented in both **2D and 3D**.

- In a **2D scatter plot**, two attributes are represented along the **X-axis and Y-axis**.
- In a **3D scatter plot**, a third attribute is represented along the **Z-axis**.
- Each point in the plot represents one **data record**.
- Scatter plots are useful for identifying **relationships and correlations between attributes**.
- They can also help identify **clusters and outliers** in the data.



2D Scatter Plot



3D Scatter Plot

Hyper Slice

Hyper Slice is a visualization technique used to represent **scalar functions involving many variables (dimensions)**.

The technique presents a multidimensional function in a **simple and easy-to-understand form**, while treating all dimensions **equally**.

The main concept of Hyper Slice is to represent a **multidimensional function as a matrix of orthogonal two-dimensional slices**.

Key Points

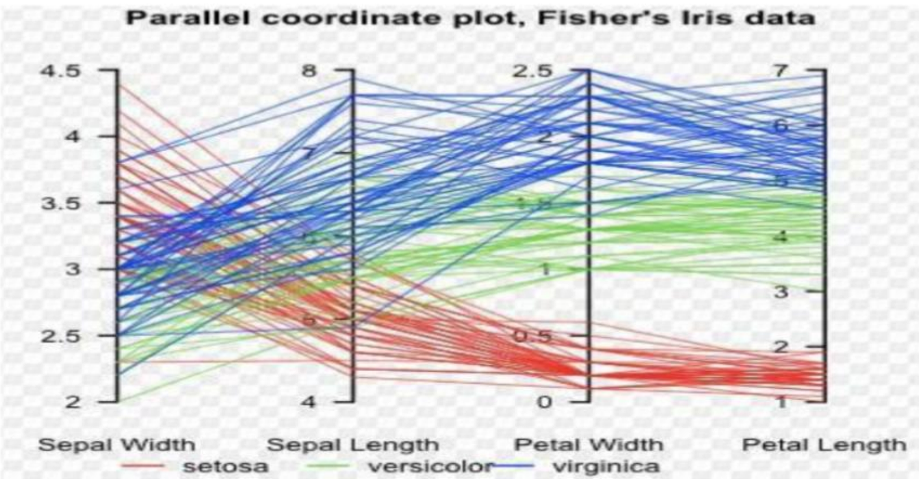
- It is designed for visualizing **multidimensional scalar functions**.
- A high-dimensional function is represented using multiple **2D slices**.
- These 2D slices are arranged in the form of a **matrix**.
- All dimensions are treated **identically**.
- The slices can be explored through **direct manipulation**.
- There is a **one-to-one relationship between screen space and variable space**, making interaction easier and more intuitive.

Parallel Coordinates

Parallel Coordinates is a geometric visualization technique used to represent n-dimensional data.

Working Principle

- To visualize **n-dimensional data**, the technique draws **n equally spaced parallel axes**.
- Each axis represents **one dimension or attribute**.
- All axes are arranged parallel to one another.
- Each data record is represented by a **polygonal line**.
- The line intersects each axis at the position corresponding to the value of that particular dimension.



Advantages

- Can represent **multiple dimensions simultaneously**.
- Useful for identifying **patterns and relationships** among attributes.
- Each data record can be traced across all dimensions.

Limitation

The major limitation is that parallel coordinates **do not work effectively with very large datasets**.

Icon-Based Visualization Techniques

Icon-based visualization techniques represent data using **icons or glyphs**. The properties of an icon are changed according to the values of different data attributes. This allows multidimensional data to be represented in a compact and visually understandable form.

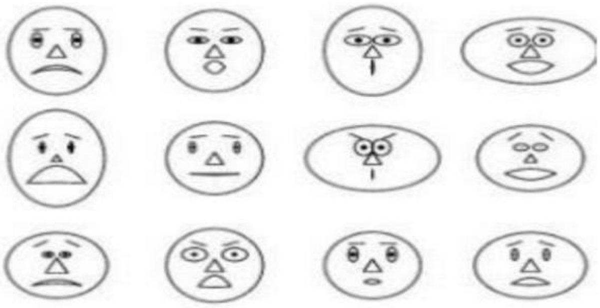
1. Chernoff Faces

Chernoff faces are an early and popular icon-based visualization technique. In this method, different data attributes are mapped to different parts of a human face, such as:

- Eyes
- Nose
- Mouth
- Face shape
- Eyebrows
- Ears

For example, the **wealth of people** can be represented using the shape of the mouth:

- **Rich people → Happy/smiling mouth**
- **Poor people → Sad/frowning mouth**



Thus, several variables can be represented simultaneously using a single face. Since humans are naturally good at recognizing differences in faces, Chernoff faces can make patterns and differences in multidimensional data easier to identify.

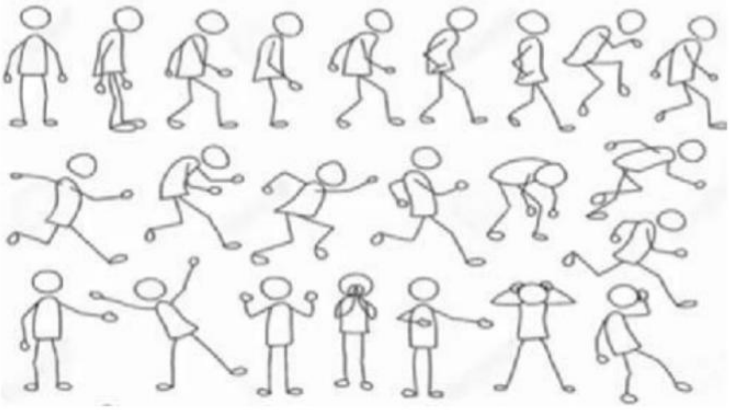
2. Stick Figures

Stick figures represent multidimensional data using a simple figure consisting of a **body and four limbs**.

In this technique:

- Two dimensions are mapped to the **X-axis and Y-axis** of the display.
- The remaining dimensions are represented by the **length and/or angle of the limbs**.
- Different combinations of limb lengths and angles produce different stick figures.

For example, if a dataset contains several attributes such as **income, age, education, and expenditure**, these attributes can be represented by different properties of the stick figure.



Advantages:

- Can represent multiple dimensions simultaneously.
- Easy to compare patterns visually.
- Useful for identifying clusters and unusual data points.

Limitation: When many dimensions are represented, the icons can become difficult to interpret.

Hierarchical Visualization Techniques

Hierarchical visualization techniques are used to represent data that has a tree-like structure or contains hierarchical relationships. These techniques organize data into different levels, making it easier to understand relationships between data items.

There are two basic approaches to hierarchical visualization:

Node-Edge Graph Layout:

This approach represents data using nodes and edges. It mainly focuses on showing the structure, connections, and relationships between different elements in the hierarchy.

Space-Filling Approach:

This approach represents hierarchical data by dividing the available display space among different nodes. It mainly focuses on showing the relative size or importance of the nodes.

Need for Hierarchical Visualization

Most visualization techniques represent multiple dimensions simultaneously. However, when the dataset is large and has high dimensionality, displaying all dimensions at the same time can make the visualization complex and difficult to understand.

Hierarchical visualization addresses this problem by:

Partitioning the dimensions into smaller subsets or subspaces.

Visualizing each subspace separately.

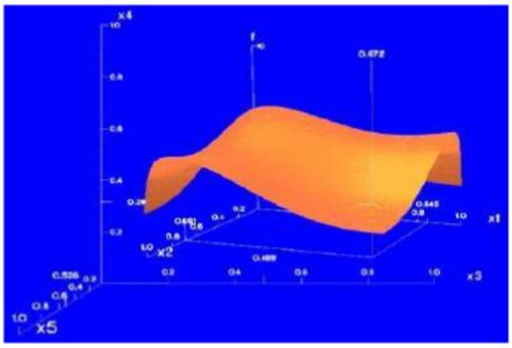
Organizing these subspaces at different hierarchical levels.

Allowing users to explore complex, high-dimensional data progressively.

Worlds-within-Worlds:

Worlds-within-Worlds, also known as n-Vision, is a representative hierarchical visualization technique.

Consider a 6-dimensional dataset with dimensions: F, X_1, X_2, X_3, X_4, X_5.



The dimensions can be divided into different groups and represented at different levels. One visualization is placed inside another, creating a hierarchy of visual spaces. When more dimensions are present, additional levels can be introduced.

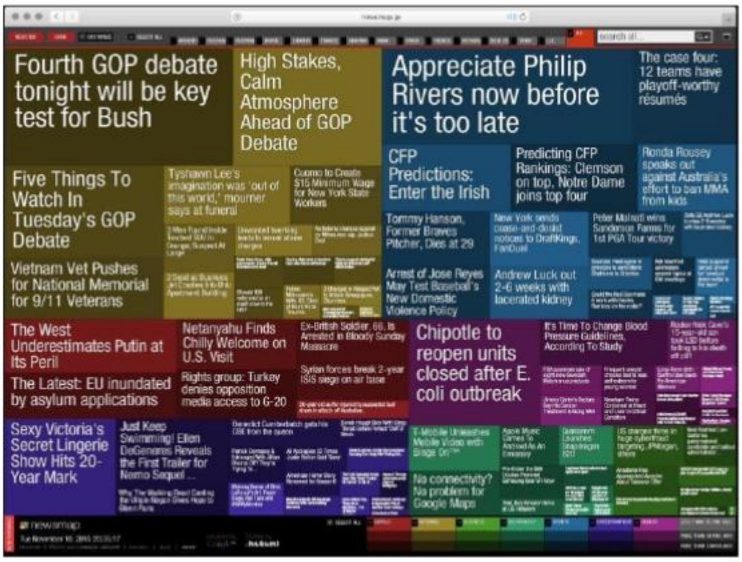
This concept of placing one visual space inside another gives the technique its name "Worlds-within-Worlds."

Tree Maps

A tree map is another important hierarchical visualization technique. It represents hierarchical data using a set of nested rectangles.

For example, in a visualization of Google News stories:

- All news stories are divided into several major categories.
- Each category is represented by a large rectangle.
- Different colors can be used to distinguish the major categories.
- Within each major category, the news stories are further divided into smaller subcategories.
- The subcategories are represented by smaller rectangles nested inside the corresponding larger rectangle.



Tree Map

Thus, a tree map provides a compact way to visualize hierarchical structure and relative sizes of data elements.

Visualizing Complex Data and Relations

Visualizing complex data and relations refers to the use of visualization techniques to represent large, multidimensional, heterogeneous, and interconnected datasets in a way that makes their patterns, relationships, and dependencies easier to understand.

1. Need for Visualizing Complex Data

Complex datasets may contain:

- A large number of records.
- Many dimensions or attributes.
- Different types of data such as numerical, categorical, textual, and temporal data.
- Relationships between different entities.
- Hierarchical and network structures.
- Frequently changing or dynamic information.

Traditional charts may not be sufficient to represent such data effectively. Therefore, specialized visualization techniques are required.

2. Visualization of Complex Relations

Relations describe how different entities or data items are connected. For example:

- People connected through social networks.
- Customers connected to products they purchase.
- Web pages connected through hyperlinks.
- Hospitals connected through patient referrals.
- Researchers connected through co-authored papers.

Graph or network visualization is commonly used to represent these relationships.

In a graph:

- **Nodes** represent entities or objects.
- **Edges** represent relationships between entities.
- **Node size, shape, or color** can represent additional attributes.
- **Edge thickness or color** can represent the strength or type of relationship.

3. Multidimensional Complex Data

When a dataset contains many dimensions, all attributes cannot always be displayed simultaneously. Techniques such as:

- Parallel coordinates
- Scatterplot matrices
- Pixel-oriented visualization
- Icon-based visualization

- Hierarchical visualization
- Geometric projection

can be used to explore different dimensions and discover patterns.

4. Interactive Visualization

Interactive visualization allows users to explore complex datasets dynamically. Common operations include:

- **Zooming:** Examining a specific portion of the visualization.
- **Filtering:** Displaying only relevant data.
- **Brushing:** Selecting a group of data items.
- **Drill-down:** Moving from summary information to detailed information.
- **Panning:** Moving across a large visualization.
- **Details-on-demand:** Displaying additional information when required.

Example:

Consider a **social network** containing thousands of users.

- Each user is represented by a **node**.
- A friendship or communication relationship is represented by an **edge**.
- Node size can represent the number of connections.
- Node color can represent the user's group or community.
- Edge thickness can indicate the frequency of communication.

Such a visualization can help identify **important users, communities, clusters, and relationships**.

Challenges

Visualizing complex data has several challenges:

1. **Overlapping:** Too many visual elements may overlap.
2. **Clutter:** Large datasets can produce crowded visualizations.
3. **High dimensionality:** Many attributes are difficult to display simultaneously.
4. **Scalability:** The visualization should remain useful as the dataset grows.
5. **Interpretability:** Users should be able to understand the displayed information easily.